AI-Based Smart Emergency Triage and Patient Flow Management System for Pakistan

1. Project Title

AI-Based Smart Emergency Triage and Patient Flow Management System for Pakistan

2. Introduction

Pakistani hospitals, especially government hospitals in cities like Karachi, Lahore, and Islamabad, face overcrowded emergency rooms (ERs). Patients are usually seen on a first-come-first-served basis rather than by how serious their condition is. This project proposes an AI agent that monitors patient vital signs, decides who needs urgent care first, and assigns doctors/beds intelligently — the same way an AI traffic system would prioritize congested lanes.

3. Problem in Pakistan

What the problem is

ERs get overloaded with patients, and staff must manually judge who to treat first, often based on who arrived earliest rather than who is most critical.

Why it's a problem in Pakistan

- Public hospitals serve very large populations with limited doctors and beds
- Manual triage is slow and inconsistent, especially during rush hours or emergencies
- Critical patients can wait behind less urgent cases

Causes

- Shortage of doctors/nurses relative to patient load
- No real-time system to track patient severity
- Paper-based or verbal patient records
- Sudden patient surges (road accidents, seasonal disease outbreaks)

How AI can help

An AI agent can continuously read vital signs (heart rate, blood pressure, oxygen level, temperature), automatically rank patients by urgency, and recommend which patient sees a doctor next — reducing wait time for critical cases and easing staff workload.

4. Proposed AI Solution

An **AI Triage Agent** that:

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- Continuously observes incoming patients' vital signs via sensors
- Classifies each patient's condition (critical / urgent / normal)
- Recommends real-time triage priority and assigns available doctors/beds
- Updates its decisions as new patients arrive or existing patients' conditions change

5. Problem Formulation

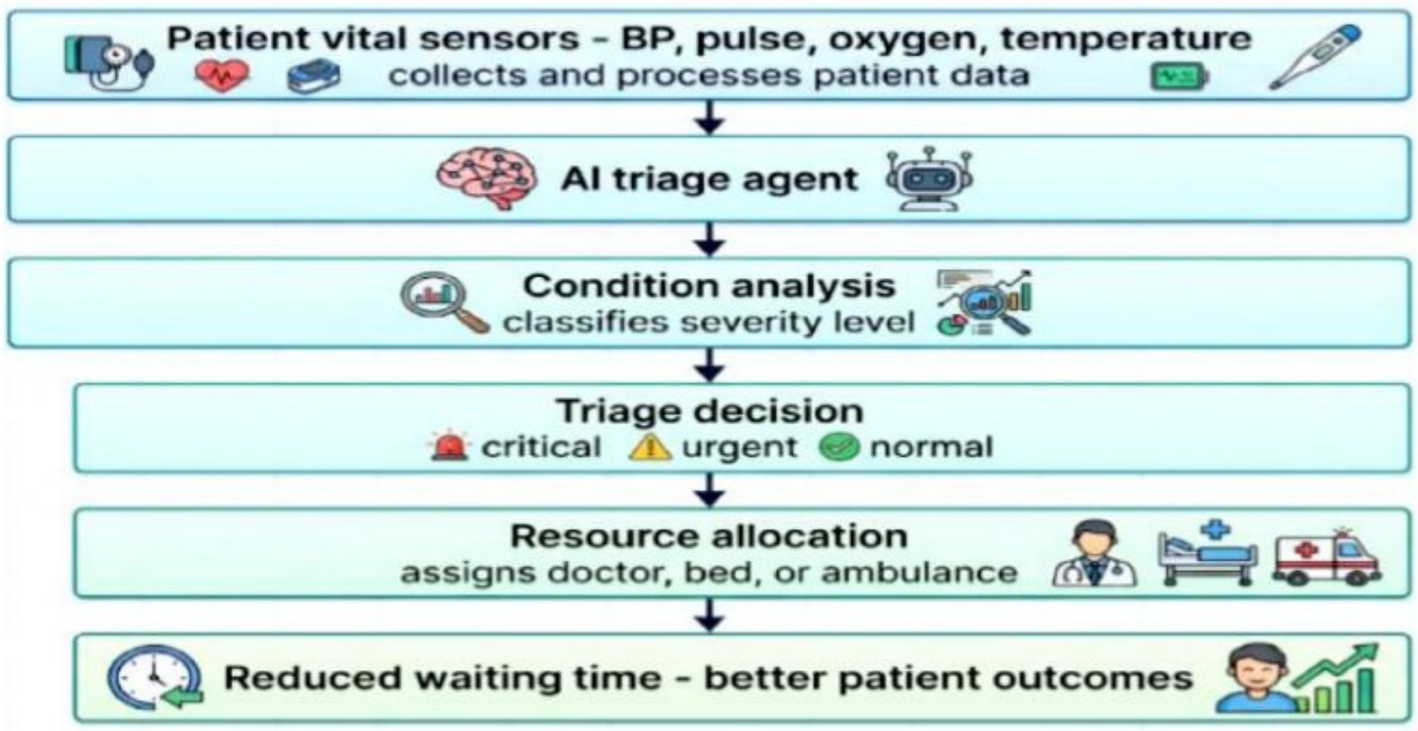
Textual formulation: Given a stream of patients arriving at an ER with varying vital signs and symptoms, decide the order and resource assignment (doctor, bed, ambulance) that minimizes waiting time for critical patients while fairly serving all patients.

- **Initial state:** ER waiting room with an initial set (or zero) patients, all resources (beds/doctors) available
- **Goal state:** All patients treated in order of medical urgency, with minimal waiting time and no critical patient left unattended
- **Actions:** Assign patient to doctor, assign bed, escalate patient to critical care, re-evaluate patient priority, place patient in queue
- **Possible states:** Different combinations of patients waiting, their severity levels, and resource availability (doctors busy/free, beds occupied/free)
- **Constraints:** Limited number of doctors and beds, limited time for critical response, unpredictable patient arrival rate, incomplete patient history info

6. Graphical Representation

The AI Triage Agent follows this flow:

Patient vital sensors → AI triage agent → condition analysis → triage decision → resource allocation → reduced waiting time



7. Environment Creation

A simple simulated environment can be built with:

- **ER waiting area:** simulated patients arriving at random times

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- **Patients:** each with randomly generated vitals (heart rate, BP, oxygen level, temperature) and a hidden 'true severity' for testing accuracy
- **Sensors:** simulated readings feeding data to the agent (can just be randomized values or a simple dataset)
- **Doctors/beds:** limited resources (e.g., 3 doctors, 5 beds) that get occupied and freed over time
- **AI triage agent:** the decision-maker

What the agent observes

Vital signs of each waiting patient, number of free doctors/beds, time each patient has been waiting.

What decisions it makes

Which patient to treat next, which doctor/bed to assign, whether to escalate a patient to critical care.

How the environment changes

Once a patient is treated, they leave the queue and free up a doctor/bed; new patients arrive continuously; waiting patients' urgency can increase over time if untreated (simulating deterioration).

8. PEAS Analysis

Component	Description
Performance measure	Minimize patient waiting time, minimize risk to critical patients, maximize number of patients treated, efficient use of doctors/beds
Environment	Hospital emergency room: patients, doctors, beds, ambulances, waiting queue
Actuators	Assigning patients to doctors/beds, updating priority queue, alerting staff for critical cases, calling for ambulance/additional resources
Sensors	Vital sign monitors (heart rate, BP, oxygen, temperature), patient check-in data, bed/doctor availability status

9. Environment Properties

Property	Classification	Reason
Observable	Partially observable	Agent only sees vitals and check-in info, not full medical history or hidden conditions
Determinism	Stochastic	Patient arrival times and how conditions evolve are unpredictable

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Property	Classification	Reason
Episodic/Sequential	Sequential	Current decisions (who gets treated first) affect future queue and outcomes
Static/Dynamic	Dynamic	New patients can arrive and conditions can worsen even while the agent is deciding
Discrete/Continuous	Continuous (with discrete decisions)	Vital signs are continuous values, but final triage categories are discrete
Single/Multi-agent	Multi-agent	Doctors, nurses, and the AI agent all act within the same environment and influence outcomes

10. Simple Working Example

A patient arrives with high heart rate and low oxygen level. Sensors send this data to the AI agent. The agent classifies the patient as 'critical', bypasses the normal queue, and immediately assigns the nearest free doctor and bed. Meanwhile, a patient with mild fever is classified 'normal' and placed later in the queue. As more patients arrive, the agent keeps re-ranking the queue in real time.

11. Problem Type (Multi - State Problem)

The proposed AI-Based Smart Emergency Triage and Patient Flow Management System can be classified primarily as a Multiple-State Problem. In this system, the AI agent must consider different possible states of the emergency room based on patients' medical severity and the availability of hospital resources.

At any given time, patients may be classified as critical, urgent, or normal, while doctors and beds may be either available or occupied. Therefore, the system has to evaluate different combinations of patient conditions, waiting queues, and resource availability before making a decision.

The AI agent continuously analyzes these possible states and determines which patient should be treated next, which doctor or bed should be assigned, and whether a patient needs to be escalated to critical care. This makes the problem a Multiple-State Problem, where the agent considers multiple possible states to achieve the goal of providing timely and appropriate medical care.

12. Conclusion

This AI-based triage system shows how intelligent agents can address real healthcare bottlenecks in Pakistan by replacing slow manual sorting with real-time, severity-based decision-making — reducing waiting time for critical patients and making better use of limited hospital resources.

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